Land Cover accuracy assessment: CEO PSUs vs ESA WorldCover (Planetary Computer) (v171)

Fix for the “sampled 0/3700 points” issue: this version (1) correctly reads the mosaic registration response key searchid/search_id, and (2) records per-request HTTP status/error so you can download the raw WorldCover extraction and audit failures.

Planetary Computer Data API reference includes mosaic endpoints and point queries (e.g., /api/data/v1/mosaic/{search_id}/point/{lon},{lat}). See the Data API reference.

Why a 3 km × 3 km AOI matters: Remote sensing products are predictions of place that always carry uncertainty tied to *spatial* resolution (pixel size and geolocation), *spectral* resolution (what wavelengths are measured), *thematic* resolution (class definitions and legend crosswalks), and *temporal* resolution (date/season and compositing choices). This tool uses a consistent **3 km × 3 km Area of Interest (AOI)** as the interpretive frame: it is large enough to represent landscape context and heterogeneity, yet small enough to connect field evidence to map predictions.

Sampling at multiple spatial resolutions: Within the 3 km AOI, land cover is summarized at multiple levels—AOI → PSU (primary sampling units) → SSU (sub-sample points). This reflects the reality that map products (e.g., 10 m WorldCover) and field sampling grids are not perfectly aligned. Comparing distributions across these nested supports makes disagreements easier to interpret (mixtures, edges, rare classes) and provides a more transparent audit trail for reconciliation.

Plot/PSU CSV: Sample/point CSV: Load (prep)

Include plot-file class labels in crosswalk list (advanced)

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1) Harmonized labels (CEO → WorldCover)

Mapping used in this report (WorldCover codes):
Build/edit the **Upload** → **WorldCover** crosswalk using pick lists. This crosswalk is applied to the uploaded SSU labels so the reference legend is harmonized to WorldCover for accuracy metrics.

Upload labels (from your CSVs)

Tip: select an upload label, choose a WorldCover class, then click **Set mapping**.

WorldCover producer definitions (legend)

Use these producer definitions to align your uploaded dot-grid elements with the WorldCover legend. Select a WorldCover class on the right to view its definition.

Select a WorldCover class…

Sources: ESA WorldCover Product User Manual (PUM) + legend summaries.

WorldCover classes

Actions

Set mapping Clear mapping

Reset to defaults, Download crosswalk CSV, Apply harmonization (compute codes)

Crosswalk table (Upload → WorldCover)

Load files to populate.

Upload label colors (raw vocabulary)

Assign explicit **colors** to your uploaded SSU labels *as they appear in the sample/point CSV* (raw-first). These colors are used in **Panel a** (SSU locations) and when **Panel b color-by → SSU label (raw)**. Download/upload the palette CSV for traceable reuse.

Reset raw label colors (hash). Download raw-label palette CSV. Upload raw-label palette CSV: Apply uploaded raw palette

Load files to populate.

Semantic bridge (θ fork): classify each PSU as `Water-dominant`, `Land-dominant`, or `Edge (near θ)` using `water_fraction` vs `θ`.

θ (water-dominant if water_fraction ≥ θ): Count wetlands as Water in θ fork

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1b) Preview of first 5 parsed SSUs (after Load)

Load files to populate.

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2) WorldCover extraction (SSU / point-level) — downloadable

WorldCover year: 2020 2021 Concurrency: Sampling endpoint: Mosaic search (register + point) Direct item (STAC lookup + item/point) Asset name: Run mode: Test: one PSU (first 100 SSUs) Full AOI (all SSUs) PSU (plotid): SSU id = PSU×100 + idx (idx=0..99 in sampleid order) Fetch WorldCover & build PSU fractions Download WC extraction (SSU points) CSV Download WC fractions (PSUs) CSV

2a) Resume from saved WorldCover extract (upload)

Upload the two CSVs previously produced by this tool to resume analysis without re-sampling: (1) SSU point extraction, and (2) PSU fractions.

SSU extraction CSV: PSU fractions CSV: Load uploaded extract & rebuild panels Exit resume mode / unlock controls Reset resume inputs

Naming conventions (recommended for traceability):
• WC2020_SSU_POINTS__AOI-036__PC-STAC__10m__YYYYMMDD.csv
• WC2020_PSU_FRACTIONS__AOI-036__PC-STAC__10m__YYYYMMDD.csv
(This tool also supports the legacy pattern it generates on download: AOI3km_lon_lat_WCYYYY_apiMode_asset_runMode_SSU_points_extraction.csv and ..._PSU_fractions.csv.)
The loader will auto-detect **AOI** and **WorldCover year** from filenames when possible.

2b) SSU location visuals — a) SSU locations (hover for SSU label), b) WorldCover labels (colors = WorldCover)

Workflow: (1) inspect uploaded labels first (panel a, upload palette), then (2) apply harmonization (crosswalk) for comparisons, then (3) sample WorldCover for producer view (panel b). Hover points to see details.

Highlight mismatches Smaller scatter points (−2px radius) **SSU plot size:**
Compact Standard Large XL XXL **Label palette:**
Uploaded palette CSV (raw labels) Hash HSL Tableau 10 Set3 Dark2 Paired Accent Pastel1 **Label encode:**
Color Symbols Both **Label symbols:**
Default (semantic) Hash shapes **Symbol map CSV:**
Load symbol map Download symbol-map template CSV Download current symbol map CSV **Fraction encode:**
Color Symbols Both **Fraction symbols:**
WorldCover shapes 1 WorldCover shapes 2 **Panel b color-by:** Harmonized reference (Upload→WorldCover) WorldCover prediction (wc_code) SSU label (raw) Match vs mismatch HTTP status (audit) Custom column… Column: **Palette (categorical):**

Hash HSL Tableau 10 Set3 Dark2 Paired Accent Pastel1 **Palette (numeric ramp):** Viridis Plasma Magma Cividis Turbo Blue→Red Red→Yellow→Green Grayscale Center ramp at 0 (diverging) **Legend mode:** Both (Raw + WorldCover) Raw upload labels Harmonized (WorldCover)

a) SSU locations — uploaded labels (upload palette; pre-harmonization view)

b) SSU scatter (color-by selector + palettes apply here)

Raw upload legend (panel a):

Harmonized legend (WorldCover producer classes):

2b color-by legend (panel b):

2c) PSU location visuals — pie charts from SSU distributions

Color PSU pies by semantic domain (θ fork ring) Narrative emphasis (tree canopy story) Uses θ + wetlands-as-water from Section 1.

Color PSU markers by: None (default pies) Uploaded top label (raw) Custom column… (numeric = ramp; string = categorical) Column: **Palette (categorical):** Hash HSL Tableau 10 Set3 Dark2 Paired Accent Pastel1 **Palette (numeric ramp):** Viridis Plasma Magma Cividis Turbo Blue→Red Red→Yellow→Green Grayscale Center ramp at 0 (diverging)
Each PSU is placed at its plot-file center (center_lon/center_lat) when available; otherwise uses the mean of SSU lon/lat. Panel c pies summarize plot-file dot-grid (reference) composition; panel d pies summarize sampled WorldCover composition. Hover a pie for details.

c) Primary Sample Units — SSU labels (pie = SSU distribution)

d) Primary Sample Units — WorldCover (pie = WC distribution)

PSU color-by legend:

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3) PSU class fraction covered by PSU

Scope follows **Run mode** and selected **PSU (plotid)** in Section 2.

a) Uploaded (Map User evidence) class fraction cover by PSU

Load plot/PSU CSV to populate.

b) WorldCover class fraction cover by PSU (from sampled WC points)

Run WorldCover extraction to populate.

c) Differences (WorldCover − Uploaded) by PSU

Δ total (L1): a distributional distance between the two PSU class-fraction vectors (Uploaded vs. WorldCover). It is computed as Δ total = $\Sigma |\Delta_i|$ across all compared classes *i*, where Δ_i = (WorldCover_i − Uploaded_i).
Interpretation: 0 means the class fractions match exactly; larger values indicate increasing compositional disagreement. This is complementary to categorical accuracy (OA/UA/PA), because it summarizes *how different the mixtures are* even when the dominant class matches.
Run WorldCover extraction to populate differences.

d) Key mismatches in PSU class fractions (summary)

Run WorldCover extraction to populate the mismatch summary.

3e) Report text — Key PSU mismatches (Uploaded vs WorldCover fractions)

Run WorldCover extraction to populate this narrative summary.

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4) Accuracy assessment (CEO reference vs WorldCover prediction)

4c) How OA, UA, PA, and F1 are computed (from the confusion matrix)

Matrix orientation: rows are **Map** classes (WorldCover prediction), and columns are **Reference** classes (CEO dot-grid labels). Each cell n_{ij} is the count of SSUs with map class *i* and reference class *j*.

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Let n_ij = confusion matrix cell (Map=i, Reference=j)
Row total (map total):   n_i+ = Σ_j n_ij
Col total (ref total):   n_+j = Σ_i n_ij
Total samples:           N    = Σ_i Σ_j n_ij
True positives (diag):   TP_i = n_ii

Overall accuracy (OA):   OA = (Σ_i TP_i) / N
User’s accuracy (UA):    UA_i = TP_i / n_i+      (commission
error = 1 − UA_i)
Producer’s accuracy (PA):PA_i = TP_i / n_+i      (omission
error   = 1 − PA_i)
F1 score:                F1_i = 2 * UA_i * PA_i / (UA_i +
PA_i)
```

UA answers: “If the map says class *i*, how often is it correct?” PA answers: “Of the reference class *i*, how often did the map capture it?” F1 summarizes the balance of UA and PA for class *i*.

4d) How to read the confusion matrix visually (quick guide)

Start with the shaded diagonal: these cells are correct classifications (Map = Reference). A stronger diagonal (larger shaded values) means higher correctness.
Scan off-diagonal hotspots: large off-diagonal values show systematic confusions (which class pairs are being mixed up).
Use UA (right column): low **UA** on a row means many mapped pixels of that class are actually other reference classes (high commission error). Visually, that row has substantial off-diagonal counts.
Use PA (bottom row): low **PA** under a column means many reference pixels of that class were mapped as something else (high omission error). Visually, that column’s mass is spread across multiple rows.

Use F1 (right column) for balance: F1 is high only when **both** UA and PA are high; it drops when either commission or omission dominates.
Keep OA in context: OA summarizes diagonal/total, but can hide minority-class problems—so interpret OA together with UA/PA/F1 and class supports.

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5) Highlighted accuracy summary (at-a-glance)

Run WorldCover extraction to populate.

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6) PSU fractional cover comparison (reference vs WorldCover) — per-class scatterplots

Each panel compares PSU dot-grid fractional cover from the plot file (x-axis) to WorldCover fractional cover from sampled SSUs (y-axis), for a single WorldCover legend class. A 1:1 line is shown. (These are continuous comparisons; UA/PA above are categorical SSU-level accuracy.)

Scatter palette: WorldCover (authoritative) Colorblind-safe High-contrast Grayscale Applies to Part 6 scatter panels. Encoding can be Color / Symbols / Both (controls at right). **Part 6 encode:** Color Symbols Both **Part 6 symbols:** WorldCover shapes 1 WorldCover shapes 2

6a) Scatterplot trends (summary across PSUs)

Run WorldCover extraction to populate.

Run WorldCover extraction to populate.

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7) AOI SketchMap — Leaflet 2×2 map pane (a–d)

Embedded 2×2 Leaflet pane (synced pan/zoom) to provide spatial context alongside the Part 6 fraction scatterplots. Pane a emphasizes user evidence; b is reference/context; c is spectral; d is model outputs.

a) User
b) Reference
c) Spectral
d) Model
AOI 3×3 km box PSU 100 m squares PSU centroid dots PSU dot size
`2` SSU points (100/PSU; heavy) SSU dot footprint (ground-based) Show only SSU footprints (hide SSU points) SSU footprint color
SSU label color, Black Ground footprint opacity
`0.18` Uncertainty opacity
`0.30` Footprint stroke color
Neutral (#111) Black SSU label color Footprint stroke opacity
`1.00` Footprint stroke width
`1.0` Footprint stroke dash
Solid Dashed (4 3) Uncertainty stroke color
Neutral (#111) Black SSU label color Uncertainty stroke opacity
`1.00` Uncertainty stroke width
`1.0` Uncertainty stroke dash
Dashed (4 3) Solid Reset SSU style SSU uncertainty circle (`uncertainty r=3.75 m`; 30%) SSU footprint: `d=5 m (r=2.5 m) ≈ 19.6 m² + uncertainty r=3.75 m` PSU squares: black Map size
Compact Standard Large XL XXL Scroll behavior
Scroll page over maps (disable wheel zoom) Enable wheel/trackpad zoom in maps
Base shading
No shading Inside shading (PSU fills) Inside AOI shading (fill AOI box) Outside PSU (within AOI, excluding PSU squares) Outside AOI (AOI → 10 km context ring)
Shading opacity

`0.18` Sync pan/zoom Dual-tick scale Zoom Zoom Zoom to AOI Zoom to 10 km ring Reset view

a) User evidence

b) Reference/context

c) Spectral

d) Model outputs

Supports: AOI = 3×3 km interpretive frame (black box) · **PSU** = 100 m primary sampling units (squares + optional centroid dots) · **SSU** = point samples within PSUs (shown when zoom ≥ 16).

N

1 km
0 0.5 km 1 km

Load plot+sample CSVs (Section 0) to populate AOI extent and overlays. If you also run WorldCover extraction, SSU points will color by returned WorldCover class; otherwise they color by mapped reference label.